

# Cost

## 1. Overview

**Cost** is one of the most practical non-functional requirements in system design.

A technically perfect system may still be a bad design if it is too expensive to build, run, maintain, or retire.

Good architects optimize for:

None

Business value / Total cost

Not just technical elegance.

## 2. Types of Cost in System Design

Cost is broader than cloud bill.

Major categories:

1. Infrastructure Cost
2. Engineering Cost
3. Operational Cost
4. Monitoring / Alerting Cost
5. Reliability Cost
6. Vendor / Licensing Cost
7. Opportunity Cost
8. Migration / Decommission Cost

## 3. Infrastructure Cost

Includes:

- Compute servers
- Databases
- Storage
- CDN
- Load balancers
- Network transfer
- Backup systems
- GPU/ML workloads

Example:

None

10 servers cost more than 2 servers

3-region deployment costs more than 1 region

## 4. Engineering Cost

More complex systems need more people.

Examples:

- Microservices require platform teams
- Multi-region systems need SRE expertise
- Distributed databases need specialists

Sometimes simplest architecture saves most money.

## 5. Operational Cost

Running production systems requires:

- On-call engineers
- Incident response
- Capacity planning
- Patching
- Security reviews
- Compliance work

Even “managed services” still need ownership.

## 6. Monitoring / Alerting Cost

More alerts  $\neq$  better.

Real-time paging for every minor issue causes:

- Alert fatigue
- Burnout
- Noise
- Unnecessary staffing cost

Choose urgency levels.

## 7. Example: Alert Trade-off

### Critical Payments Failure

Need immediate paging.

### Nightly Analytics Delay

Email/slack alert next morning enough.

Different SLA = different cost.

## 8. Cost Trade-offs with Other NFRs

### A. Cost vs Availability

High availability costs more.

#### Cheap:

None

1 server

1 DB

No replica

## Expensive:

None

Multiple servers

DB replicas

Multi-AZ

Automatic failover

## B. Cost vs Performance

Lower latency often needs:

- More regions
- Caches
- Extra replicas
- CDN
- Faster hardware

## C. Cost vs Complexity

Sometimes paying more reduces complexity.

Example:

Use managed database instead of self-hosting.

Higher bill, lower ops burden.

## D. Cost vs Scalability

Overprovisioning handles growth but wastes money.

Underprovisioning saves cost but risks outages.

## 9. Vertical vs Horizontal Scaling Cost

### Vertical Scaling

Bigger machine:

None

1 powerful server



Pros:

- Simpler

Cons:

- Hardware expensive
- Single point risk

## Horizontal Scaling

Many smaller servers.

Pros:

- Better scale/fault tolerance

Cons:

- More complexity

## 10. Region / Data Center Cost

Closer regions to users reduce latency but may cost more.

Cheaper region farther away may increase latency.

Example:

None

Asia users served from US region

Cheap maybe

Slow definitely

## 11. Managed Service vs Self-Hosted

| Option         | Infra Cost   | Engineering Cost | Reliability  |
|----------------|--------------|------------------|--------------|
| Self-hosted DB | Lower direct | Higher ops       | Depends      |
| Managed DB     | Higher bill  | Lower ops        | Often strong |

Need total cost view.

## 12. Hidden Costs Engineers Miss

- Data transfer egress fees
- Backups / snapshots
- Logs storage
- Idle environments
- CI runners
- Staging clusters
- TLS cert management
- Disaster recovery infra

## 13. Cost of Technical Debt

Cheap shortcut today can cost heavily later.

Examples:

- No tests
- Hardcoded configs
- Monolithic spaghetti code
- Manual deploys

Leads to slow future development.

## 14. Dependency Lifecycle Cost

Libraries/services get deprecated.

Need upgrades.

Questions:

None

What if library unsupported?

What if vendor shuts down?

What if API pricing changes?

Third-party dependencies can become expensive later.

## 15. Vendor Lock-in Cost

Using one vendor deeply may create migration pain.

Examples:

- Proprietary DB features
- Cloud-specific queues
- Closed analytics platform

Need abstraction where sensible.

## 16. Build vs Buy Decision

Should we build internal solution or buy SaaS?

### Buy

- Faster launch
- Less engineering effort

### Build

- More control
- Potential lower long-term cost

Depends on scale and strategic importance.

## 17. Cost of Downtime

Downtime cost may exceed infra savings.

Examples:

- Lost orders
- SLA penalties
- Reputation damage
- Support load

So cheapest system is not always lowest-cost system.

## **18. Decommissioning Cost (Often Ignored)**

Systems may be retired because:

- Low adoption
- Strategy change
- Merge into new platform

Need plan for:

- Export user data
- Shut infra safely
- Archive logs
- Remove secrets/access
- Notify users

## **19. Example: Export User Data**

Provide:

- CSV
- JSON
- Account archives

Before shutdown.

## 20. Interview Questions to Ask

1. Startup or enterprise budget?
2. Cost optimization priority vs speed?
3. Peak traffic or stable traffic?
4. Need 24x7 HA?
5. Managed services allowed?
6. Forecasted growth?

## 21. Cost Optimization Techniques

- Autoscaling
- Spot/preemptible instances
- Right-sizing machines
- Tiered storage
- Cache hot reads
- Archive old data
- Use serverless for bursty workloads
- Delete idle environments

## 22. Example Answer

**Q: How would you reduce cost of this design?**

Answer:

I'd evaluate whether multi-region is necessary now, use autoscaling, move infrequent workloads to async jobs, archive cold data to cheaper storage, and prefer managed services where operational headcount savings outweigh service pricing.

## 23. Best Practices

- Design for current needs + growth path
- Measure total cost, not just compute cost
- Match SLA to business value
- Avoid overengineering
- Review vendor dependencies yearly
- Track unit economics (cost per user/order)

## 24. Trade-off Table

| Decision                    | Lower Cost        | Higher Cost           |
|-----------------------------|-------------------|-----------------------|
| Single region               | Yes               | No                    |
| Multi-region HA             | No                | Yes                   |
| No replicas                 | Yes               | Risky                 |
| Managed DB                  | Maybe higher bill | Lower ops cost        |
| Real-time alerts everywhere | Expensive         | Sometimes unnecessary |
| Approx analytics            | Cheaper           | Exact may cost more   |



## 25. Short Revision Notes

None

Cost includes:

- Infra
- Engineers
- Operations
- Monitoring
- Downtime
- Vendor lock-in
- Decommissioning

Trade-offs:

Higher HA/performance usually costs more

Simpler systems often cheaper overall